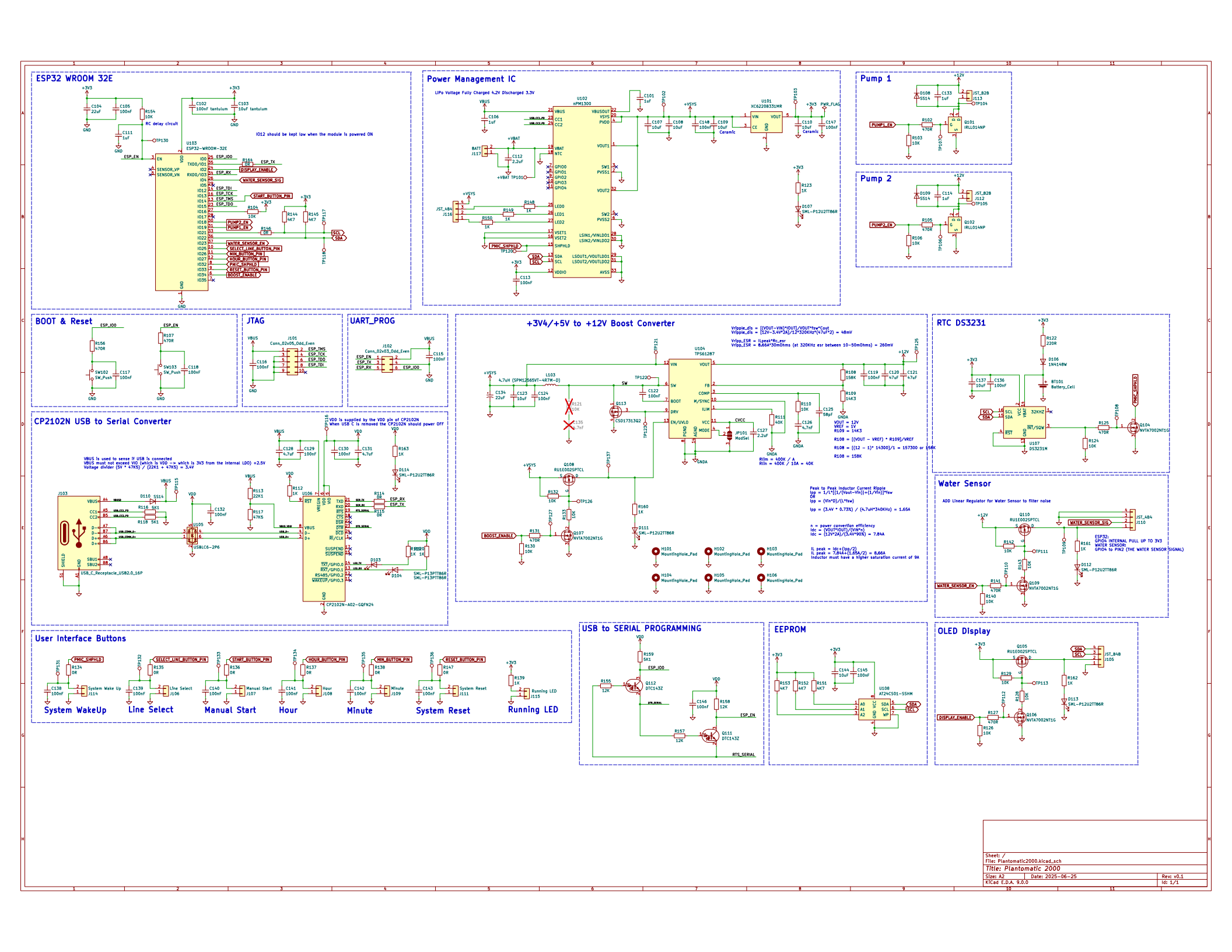

ESP32 WROOM 32E
IO12 should be kept low when the module is powered ON
RC delay circuit
Power Management IC
LiPo Voltage Fully Charged 4.2V Discharged 3.3V
Pump 1
Pump 2
BOOT & Reset
JTAG
UART_PROG
+3V4/+5V to +12V Boost Converter
Vripple_dis = [(VOUT-VIN)*IOUT]/VOUT*fsw*Cout
Vripple_dis = [12V-3.4V*2A]/12*320KHz*(47uF*2) = 48mV
Vrip_ESR = Ilpeak*Rc_esr
Vrip_ESR = 8.66A*30mOhms (at 320KHz esr between 10-50mOhms) = 260mV
R108 = [(VOUT - VREF) * R109]/VREF
R108 = [(12 - 1)* 14300]/1 = 157300 or 158K
R108 = 158K
Rilm = 400K / A
Rilm = 400K / 10A = 40K
Peak to Peak Inductor Current Ripple
Ipp = 1/L*[(1/(Vout-Vin))+(1/Vin)]*fsw
OR
Ipp = (Vin*D)/(L*fsw)
Ipp = (3.4V * 0.73%) / (4.7uH*340KHz) = 1.65A
n = power convertion efficiency
Idc = [VOUT*IOUT]/(Vin*n)
Idc = [12V*2A]/(3.4V*90%) = 7.84A
IL peak = Idc+(Ipp/2)
IL peak = 7.84A+(1.65A/2) = 8.66A
Inductor must have a higher saturation current of 9A
RTC DS3231
Water Sensor
ADD Linear Regulator for Water Sensor to filter noise
ESP32: GPIO4 INTERNAL PULL UP TO 3V3
WATER SENSOR: GPIO4 to PIN2 (THE WATER SENSOR SIGNAL)
CP2102N USB to Serial Converter
VBUS is used to sense if USB is connected
VBUS must not exceed VIO (which is VDD -> which is 3V3 from the internal LDO) +2.5V
Voltage divider (5V * 47K5) / (22K1 + 47K5) = 3.4V
VDD is supplied by the VDD pin of CP2102N
When USB C is removed the CP2102N should power OFF
User Interface Buttons
System WakeUp
Line Select
Manual Start
Hour
Minute
System Reset
Running LED
USB to SERIAL PROGRAMMING
EEPROM
OLED Display
Sheet: /
File: Plantomatic2000.kicad_sch
Title: Plantomatic 2000
Size: A2
Date: 2025-06-25
Rev: v0.1
KiCad E.D.A. 9.0.0
Id: 1/1